Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2021
Department of Materials Science and Engineering

EEE 1127
(Electrical Engineering Fundamentals)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A
(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Define electrical circuit and its parameters. A stove element draws 15A when connected to a 240V. How long does it take to consume 180kJ. (08)

(b) Find V_o and the power absorbed by each element in the circuit of Fig.Q1(b) (10)

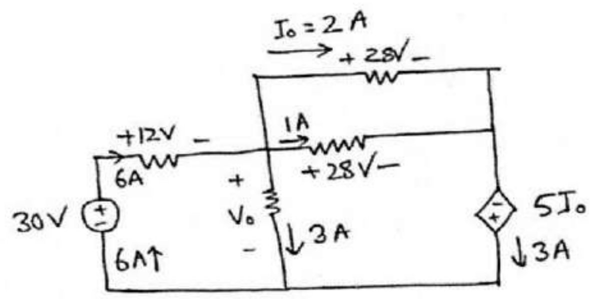

Figure: 1(b)

(c) State and explain Kirchoff's Voltage Law (KVL) and Kirchoff's Current Law (KCL) and find the mesh currents for the circuit as shown in Fig.Q1(c). (10)

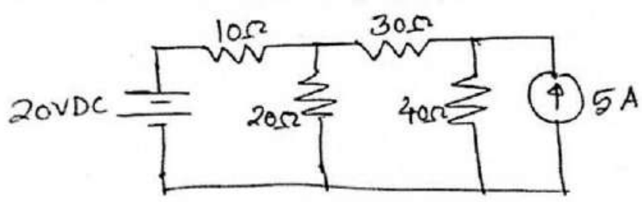

Figure: 1(c)

(d) Deduce the expressions for delta to wye ($\Delta - Y$) conversion. (07)

Q 2. (a) Explain the steps of Nodal analysis. Apply nodal analysis to find i_o and the power dissipated in each resistor in the circuit of Fig.Q2(a). (12)

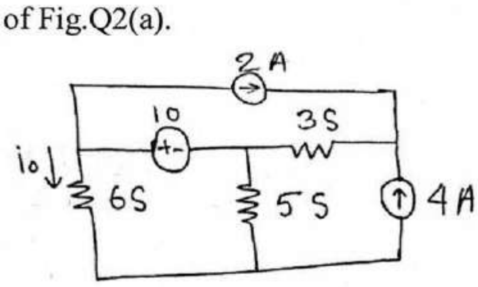

Figure: 2(a)

(b) Use mesh analysis to find V_{ab} and i_o in the circuit of Fig.Q2(b). (12)

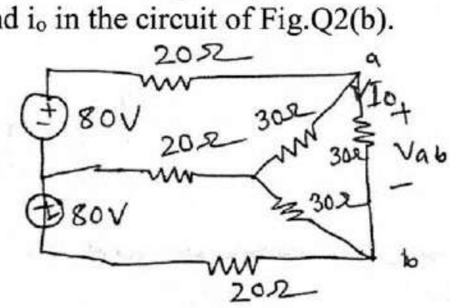

Figure: 2(b)

(c) Use source transformation to find the voltage V_x in the circuit of Fig.Q2(c). (11)

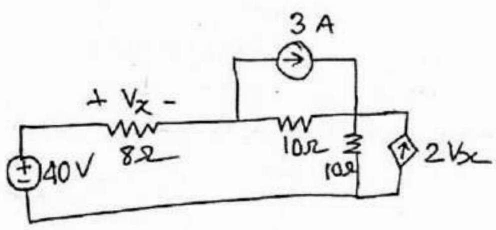

Figure: 2(c)

Q 3. (a) State and explain superposition theorem. Apply superposition theorem in the circuit as shown in Fig.Q3(a) and find out the currents I_1, I_2, I_3 and I_4. (13)

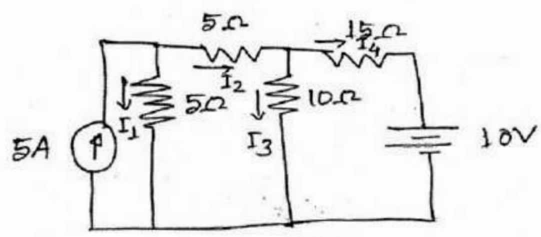

Figure: 3(a)

(b) Find the thevenin and Norton equivalent circuits at the a-b terminal for the circuit in Fig.Q3(b). (12)

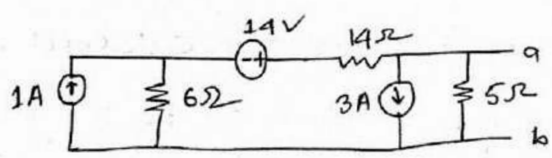

Figure: 3(b)

(c) Find current i_2 for the following network as shown in Fig.Q3(c). (10)

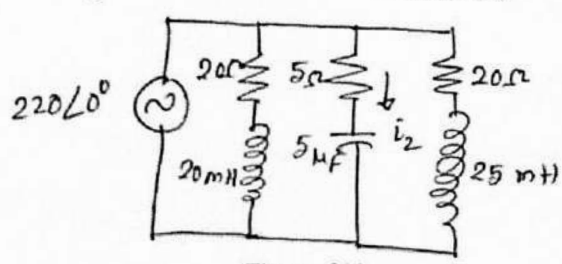

Figure: 3(c)

Q 4. (a) Calculate the phase angle between $v_1 = -10\sin(wt + 50°)$ and $v_2 = 12\sin(wt - 10)$. Determine which one is leading? (10)

(b) Express these sinusoidal as phases: (08)

$v = 7cos(2t + 40°)$ & $i = -4\sin(10t + 10°)$. Also, find the resultant $P = Vi$.

(c) Convert $\log_e(\sqrt{\frac{150\angle-90°}{5\angle 90°}})$. (06)

(d) Define amplitude, phase, period and frequency. Find these values from (11)

$v(t) = -10\sin(10t + 12°)$. Also, show $v(t)$ in the graph with approximate scaling,

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Describe the function of i) commutator ii) Armature iii) Pole core iv) Pole shoe and v) Yoke. (08)

(b) For a locomotive, which DC generator do you recommend and why? (07)

(c) Derive the condition for maximum efficiency in generator. Therefore, find the load current corresponding to maximum efficiency. (10)

(d) A short shunt DC generator supplies 250V at 100A current in a load. The generator has armature, series-field, and shunt-field resistances are 0.05Ω, 0.3Ω and 200 Ω respectively. If brush contact drop and armature reaction drop is 1.0V per brush and 3V respectively, find out the generated voltage of generator. (10)

Q 6. (a) Classify different types of motor. How does a DC motor produce unidirectional torque through commutation? Explain with details schematics. (15)

(b) Draw and explain the different characteristic curves of DC shunt and series motor. For a conveyer belt application in industry which motor do you recommend and why? (10)

(c) Describe the effect of back emf and armature reaction. Also, derive the condition of maximum power in a DC motor. (10)

Q 7. (a) Why transformer is necessary in power transmission? Explain mathematically. Describe different types of cooling methods of transformer. (10)

(b) Describe the operating principle of a 1- φ transformer. Also, derive the expression for induced emf in any of the side. (12)

(c) Define all day efficiency of transformer. Design a step down transformer where the HV side has 11KV and LV side has x volt (x= last two digit of your roll number). To operate the transformer at maximum efficiency what will be the secondary current of your designed transformer? Assume that the core loss is 300 watt and equivalent resistance with respect to primary is 200 Ω. (13)

Q 8. (a) Describe the working principle of induction motor. Differentiate between DC motor and induction motor. (10)

(b) Show that 3-phase supply in the stator of an induction motor can create a revolving field with constant magnitude. (12)

(c) Describe the advantages of Alternator over DC generator. (05)

(d) Describe the operation of an alternator. (08)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2021
Department of Materials Science and Engineering

MSE 1101

(Introduction to Materials Science and Engineering)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Why metals are ductile but ceramics are brittle? (05)

(b) Distinguish between the following properties of materials: (10)

(i) Strength and Hardness
(ii) Modulus of elasticity and Modulus of resilience
(iii) Stiffness and Toughness

(c) Draw a neat sketch of a body-centered cubic unit cell and determine its (20)

(i) Lattice parameter (a) in terms of atomic radius (r)
(ii) Co-ordination number
(iii) Number of atoms per unit cell
(iv) Atomic packing density

Q 2. (a) Differentiate between crystal and lattice. (05)

(b) With the help of appropriate diagrams, explain the following: (15)

(i) The dependence of repulsive, attractive and net forces on interatomic separation for two atoms.
(ii) The dependence repulsive, attractive and net potential energies on interatomic separation for two isolated atoms.

(c) Compute the radius r of an impurity atom that just fits into a FCC tetrahedral and octahedral sites in terms of the atomic radius R of the host atom without introducing lattice strains. (15)

Q 3. (a) Why are defects important in materials? Explain Schottky and Frenkel defect with neat sketches. (10)

(b) Calculate the activation energy for vacancy formation in aluminum (Al), given that the equilibrium number of vacancies at 500° C is 7.57×10^{23} m^{-3}. The atomic weight and density (at 500° C) for aluminum (Al) are 26.98g/mol and 2.62 g/cm^3, respectively. (10)

(c) Define fatigue. What will be appearance of a fatigue fracture surface? Describe the factors that affect fatigue life. (15)

Q 4. (a) What is true stress? Compare true stress-strain behavior with engineering stress-strain behavior with the help of necessary figure. (05)

(b) Table 4 (b) (10)

Material	*Yield Strength (MPa)*	*Tensile Strength (MPa)*	*Strain at Fracture*	*Fracture Strength (MPa)*	*Elastic Modulus (GPa)*
A	310	340	0.23	265	210
B	100	120	0.40	105	150
C	415	550	0.15	500	310
D	700	850	0.14	720	210
E	Fractures before yielding			650	350

(i) Which will experience the greatest percentage reduction in area? Why?
(ii) Which is the strongest? Why?

(iii) Which is the stiffest? Why?

(c) Figure 4 (c) from the tensile stress-strain behavior for the brass specimen shown in figure, determine the following: (20)

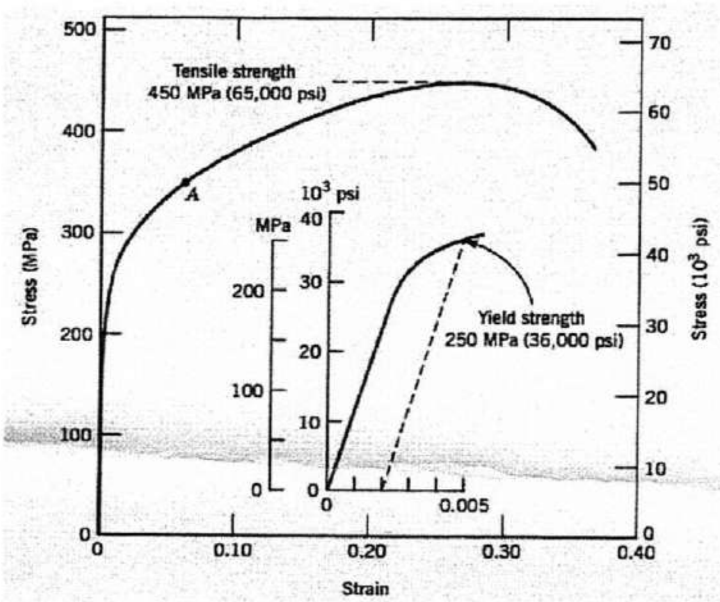

Figure 4 (c) The stress-strain behavior for brass specimen

(i) The modulus of elasticity

(ii) The yield strength at a strain offset of 0.002

(iii) The maximum load that can be sustained by a cylindrical specimen having an original diameter of 12.8 mm (0.505 in.)

(iv) The change in length of a specimen originally 250 mm (10 in.) long that is subjected to a tensile stress of 345 MPa (50,000 psi).

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) What is composite materials? Explain, why length and diameter of fibers play the vital roles for the enhancement of load bearing capability of composite materials with detail sketches. (10)

(b) Mention different manufacturing methods that are used for the production of reinforced thermoplastic and reinforced thermoset composites. (05)

(c) Deduce an expression for the modulus of elasticity in its fiber direction for a hybrid composite in which all fiber of both types are oriented in the same direction. Suppose this hybrid composite consists of aramid and glass fiber having volume fraction of 0.25 and 0.35, respectively, with a polyester resin matrix. The modulus of elasticity of aramid fiber, glass fiber and matrix is 125 GPa, 69 GPa and 4 GPa, respectively. (20)

(i) Now compute the of modulus of elasticity of this composite in the longitudinal direction and in the transverse direction.

Q 6. (a) Mention various methods to strengthen metal. Explain the solid solution strengthening mechanism with proper example. (10)

(b) Briefly explain the effect of aging temperature and time on precipitation hardening of Al alloy. (10)

(c) List the different technique methods of NDT. Describe any one NDT method to detect materials internal defects using necessary figures. (15)

Q 7. (a) Explain, why no hole is generated by the electron excitation involving a donor impurity atom (10)

(b) The room temperature electrical conductivity of a silicon specimen is 500 $(\Omega.m)^{-1}$. The hole concentration is known to be 2.0×10^{22} m^{-3}. Using the electron and hole mobilities for silicon in the following table 7 (b). (10)

(i) Compute the electron concentration

(ii) On the basis of result in part (i), is the specimen intrinsic, n-type extrinsic on p-type extrinsic? Why explain.

Material	Band Gap (eV)	Electron mobility (m^2/V-s)	Hole mobility (m^2/V-s)	Electrical conductivity (intrinsic) $(\Omega.m)^{-1}$
Ge	0.670	0.390	0.190	2.200
Si	1.110	0.145	0.050	3.4×10^{-4}

(c) What is electron energy band structure and fermi energy? Classify materials on the basis of electron band structure. (09)

(d) Differentiate between Neel and Curie temperature. (06)

Q 8. (a) Briefly discuss the optical phenomena within solid when light interest with solid. (10)

(b) What is antiferromagnetism? How does temperature affect the magnetic behavior of antiferromagnetic material? Explain. (10)

(c) What is photoconductivity? Briefly discuss the construction and working principle of solar cell with a neat sketch. (15)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2021
Department of Materials Science and Engineering

Ch 1127
(Inorganic and Physical Chemistry)

Time: 3 hours — Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A
(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) Draw the Lewis structures for the following species: i) NO_3^- ii) SO_4^{2-} iii) HNO_3 iv) O_3. (08)

(b) An ionic bond is formed between a cation A^+ and an anion B^-. How would the bond be affected by the change of i) doubling the ionic radius of A^+ ii) tripling the charge on A^+. (11)

(c) Explain different types of hydrogen bonding. Discuss the role of hydrogen bonding on melting and boiling points of organic and inorganic compounds. (09)

(d) Use the following data to calculate the lattice energy of calcium oxide. You must write all thermochemical equations for the steps of the cycle. (07)

The enthalpy of formation of CaO(s) = -636 kJ/mole
The enthalpy of sublimation of Ca = +192 kJ/mole
1st ionization energy of Ca = +590kJ/mole
2nd ionization energy of Ca = +1145kJ/mole
The enthalpy of dissociation of O_2 = +494 kJ/mole
1st electron affinity of O(g) = -141 kJ/mole
2nd electron affinity of O(g) = +845 kJ/mole

Q 2. (a) Explain an electrochemical cell of Galvanic type and electrolytic type. Write down the sign convention and types of reaction of these cells. (10)

(b) Derive an expression for Nernst equation. (10)

(c) Determine the feasibility of the reaction (06)

$Al(s) + 3Sn^{4+}(aq) \rightarrow 2Al^{3+}(aq) + 3Sn^{2+}(aq)$

Given, $E^\circ_{Al/Al^{3+}} = -1.66V$ and $E^\circ_{Sn^{4+}/Sn^{2+}} = 0.15V$

(d) What is meant by ionic mobility, Faraday and Standard electrode potential? (09)

Q 3. (a) What is liquid junction potential? How can it be removed? (05)

(b) Is it possible to store copper sulphate in an iron vessel for a long time? (06)

Given : $E_{Cu^{2+}/Cu} = 0.34V$ and $E_{Fe^{2+}/Fe} = -0.44\,V$

(c) "On progressive dilution, specific conductance of an electrolyte decrease but molar conductance increases"- discuss. (12)

(d) What is meant by the transport number of an ion? How is it determined? (12)

Q 4. (a) What is order of a reaction? Discuss one method for determining the order of a reaction. (10)

(b) What is the half-life of a reaction? Show that for a first order reaction half-life is independent of initial concentration. (08)

(c) When a mixture of methane and bromine is exposed to light, the following reaction occurs slowly: $CH_4(g) + Br_2(g) \rightarrow CH_3Br(g) + HBr(g)$ (07)

(d) Derive rate equation for second order reaction when the reactants are same. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Give one example of each: a gaseous solution, a liquid solution, a solid solution. (06)

(b) Explain in terms of intermolecular interactions why octane is immisible in water. (08)

(c) Discuss about supersaturated solution with suitable example. (10)

(d) State Raoult law and obtain expression for lowering of vapour pressure when non-volatile solute is dissolved in solvent. (11)

Q 6. (a) What is the way of protecting an emulsion from coalescence? (08)

(b) What is electrophoresis? How does this phenomenon provide information about the sign of charge on particles? (12)

(c) Write down the application of colloids in artificial kidney machine. (09)

(d) What distinguishes sol from gel? (06)

Q 7. (a) What is "phase diagram"? Explain i) triple point ii) eutectic point with reference to phase diagram. (10)

(b) Explain the phase diagram of two components system having congruent melting point. (10)

(c) An alloy of Cd and Bi contains 25% Cd. Find the mass of eutectic in 1Kg of the alloy, if the eutectic systems contains 40% Cd. (09)

(d) Find out the number of degree of freedom in the following systems: (06)

i) Sulphur(l) $\rightleftharpoons$ Sulphur(vap) ii) Saturated solution of NaCl

iii) $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$

Q 8. (a) Define an acid and a base according to the Lewis concept. Give a chemical reaction to illustrate. (10)

(b) Write an equation in which $H_2PO_3^-$ acts as an acid and another in which it acts as a base. (08)

(c) Give the conjugated acid to each of the following species regarded as bases. (10)

i) ClO^- ii) TeO_3^{2-} iii) NH_2^- iv) N_2H_4 v) HS^-

(d) A shampoo solution at 25°C has a hydroxide ion concentration of 1.5×10^{-9} M. Is the solution acidic, neutral or basic? (07)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2021
Department of Materials Science and Engineering

Ph 1127
(Optics and Waves)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q 1. (a) What are the properties of simple harmonic motion? Show that the kinetic energy and potential energy of a particle executing SHM is dependent on time, while the total energy is not. (12)

(b) Discuss damped harmonic motion and show that it is possible to get damped harmonic oscillation as an oscillatory motion. (13)

(c) A body is vibrating with simple harmonic motion of amplitude 15 cm and frequency 4 Hz. Calculate (10)

(i) The maximum value of the acceleration and velocity, and
(ii) The acceleration and velocity when the displacement is 9 cm.

Q 2. (a) Write down the characteristics of wave motion. Distinguish the travelling and stationary wave. (10)

(b) Define antinodes and nodes. Given that, $y = A \sin\frac{2\pi}{\lambda} vt$ is the equation of standing wave, and when a longitudinal wave propagates through a fluid, the bulk modulus of the fluid is (15)

$k = -\frac{P}{dy/dx}$. Now show that no energy is transfer in case of standing waves.

(c) A source of sound has a frequency of 256 Hz and an amplitude of 5×10^{-3}m. What is the flow of energy across a square meter per second if the velocity of sound in air is 350 ms^{-1} and density of air is 1.26 kgm^{-3}. (10)

Q 3. (a) Explain Doppler's effect. A source produces a note of frequency 'f' and is moving towards a satisfactory observer with a uniform speed 'a'. Show that apparent pitch is, (13)

$f' = f(\frac{V}{V-a})$, 'V' is the velocity of sound in air.

(b) Write some applications of ultrasonic waves. (11)

(c) The apparent frequency of the whistle of an engine changes in the ratio 8:4 as the engine passes a stationary observer. If the velocity of the sound is 342 ms^{-1}, calculate the velocity of engine. (11)

Q 4. (a) What is the difference between Phon and Bel. Discuss how energy density and Intensity of sound are related. (15)

(b) What is meant by acoustic intensity level and acoustic pressure level. (10)

(c) Calculate the increase in the acoustic intensity level when the sound intensity is double. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) Show that, the equivalent focal length of two thin lenses separated by a finite distance is equal to $f = \frac{f_1 f_2}{f_1+f_2-d}$, Where symbols have their usual meaning. (13)

(b) Draw a suitable ray diagram of compound microscope and hence show that the magnifying power of compound microscope is, (12)

$M = \left(1+\frac{D}{f_e}\right) \times \left({V}/{f_o} - 1\right)$, Where symbols have their usual meaning.

(c) The glass have a 2:3 ratio of dispersive powers. These glasses is used to make an achromatic objective with 20 cm focal length. What are the focal lengths of the lenses? (10)

Q 6. (a) Why the Newton's rings are circular and how does interference take place in Newton's rings experiment? (10)

(b) Superposition principle is the basic principle used in the interference phenomena. Let at any time t, the displacement of wave at point p from slit S_1 is y_1 and from slit S_2 is y_2. (15)

(i) If the incoming light waves superimpose constructively, then show that, path difference = $n\lambda$.

(ii) If the incoming light waves add destructively, then show that, path difference = $(2n+1)\lambda/_2$.

(iii) Hence show that the separation between two consecutive bright or two consecutive dark fringe are same.

(c) The diameter of the 9th dark ring in a Newton's Ring system viewed normally by reflected light of wavelength 5800 Å. Calculated the radius of curvature of the lens and the thickness of the air film. (10)

Q 7. (a) What is meant by diffraction of light? Write down the basic differences between Fresnel and Fraunhofer diffraction. (10)

(b) Describe and explain the phenomenon of diffraction due to a straight edge. Explain why the bands are neither equidistance nor equally illuminated. (15)

(c) In Fraunhofer diffraction pattern due to a single slit, the screen is at a distance of 100 cm from slit of width 0.1 mm. The slit is illuminated by monochromatic light of wavelength 5800 Å. Calculate the distance of 2nd minimum from the central maximum. (10)

Q 8. (a) What are positive and negative crystal? Explain polarization of light on the basis of electromagnetic theory. (10)

(b) What is double refraction? Discuss the basic construction of Nicol Prism. Hence Explain how the unpolarized light on passing through the Nicol Prism becomes plane polarized light. (15)

(c) When a 20 cm long tube carrying 48 cm^3 cube of sugar solution is put in a saccharimeter, it creates an optical rotation of 22°. Calculate the amount of sugar in the tube in the form of a solution if the specific rotation of sugar solution is 66°. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term Examination, 2021
Department of Materials Science and Engineering

Math 1127
(Differential and Integral Calculus)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

Q1. (a) A function $f(x)$ is defined by as follows; (13)

$$f(x) = \begin{cases} x\,; & 0 < x < 1 \\ 2 - x\,; & 1 \leq x \leq 2 \\ 2x - x^2\,; & x > 2 \end{cases}$$

Discuss the continuity of $f(x)$ at $x = 1$ and differentiability of $f(x)$ at $x = 2$.

(b) Evaluate $\lim_{x \to 0} (\frac{tanx}{x})^{1/x^2}$. (10)

(c) If $y = \sin^{-1}(\frac{2x}{1+x^2})$, then find y_n. (12)

Q2. (a) Define maxima of a function at a point. Find the extreme value of the function $f(x) = \sin x\,(1 + \cos x)$ in the interval $0 \leq x \leq \pi$. (13)

(b) State Euler theorem on homogeneous function and verify it for the function (12)

$u = x^2 f(y/x)$.

(c) Expand $\ln(\sin x)$ in the powers of $(x - 3)$. (10)

Q3. (a) Show that the maximum value of $x^2 \log(\frac{1}{x})$ is $\frac{1}{2e}$. (11)

(b) If $u = F\,(x^2 + y^2 + z^2) f(xy + yz + zx)$. (12)
Prove that, $(y - z)\frac{du}{dx} + (z - x)\frac{du}{dy} + (x - y)\frac{du}{dz} = 0$.

(c) Find the tangent and the normal at the point $(1, -1)$ to the curve (12)
$x^3 + xy^2 - 3x^2 + 4x + 5y + 2 = 0$.

Q4. (a) Find all possible asymptotes for the curve $(y + 3)(x^2 - 4x + 3) + 2x - 3 = 0$. (10)

(b) If $u = y^2 f(x/y) + y\psi(x/y) + \phi(x/y)$, (13)
then prove that $x^2 \frac{d^2u}{dn^2} + 2xy\, \frac{d^2u}{dndy} + y^2 \frac{d^2u}{dy^2} = 2y^2 f(x/y)$.

(c) Find the radius of curvature and center of circular of curvature of the ellipse (12)
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(0, b)$.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q5. (a) Integrate the followings : (12)

$\int (x+1)\sqrt{x^2-2x+5}\,dx.$

(b) $\displaystyle\int \frac{dx}{(x+1)\sqrt{1+2x+x^2}}.$ (13)

(c) $\displaystyle\int \frac{5sinx+4cosx}{3cosx+sinx}dx.$ (10)

Q6. (a) Evaluate, $\int_0^{\infty} \log(x+\frac{1}{x})\frac{dx}{1+x^2}.$ (15)

(b) Evaluate, $\lim\limits_{n\to\infty}[\frac{1^2}{n^3+1^3}+\frac{2^2}{n^3+2^3}+\cdots+\frac{1}{2n}].$ (10)

(c) Evaluate, $\int_0^{\pi} \frac{xtanx}{secx+tanx}dx.$ (10)

Q7. (a) Define Beta function and Gamma function. Use Gamma function to evaluate $\int_0^{\pi} cos^4xdx.$ (12)

(b) Form a reduction formula for $\int tan^n xdx$ and hence evaluate $\int tan^5 xdx.$ (11)

(c) Evaluate the double integral $\iint_R xydxdy$, where R is the region bounded by the x-axis, the line $y=2x$ and the parabola $x^2=4ay.$ (12)

Q8. (a) Determine the area of the parabola $y^2=2x$ intercepted by the line $y=4x-1.$ (12)

(b) Find the area inside the circle $r=\sin\theta$ and outside the cardioid $r=1-\cos\theta.$ (13)

(c) Evaluate, $\int_0^1\int_x^{x^2}\int_{xy}^{x^2y^2} xydzdydx.$ (11)